6.4.1 Numerical stability of triangular solve

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Week 06 due Feb 21, 2025 11:05 IST

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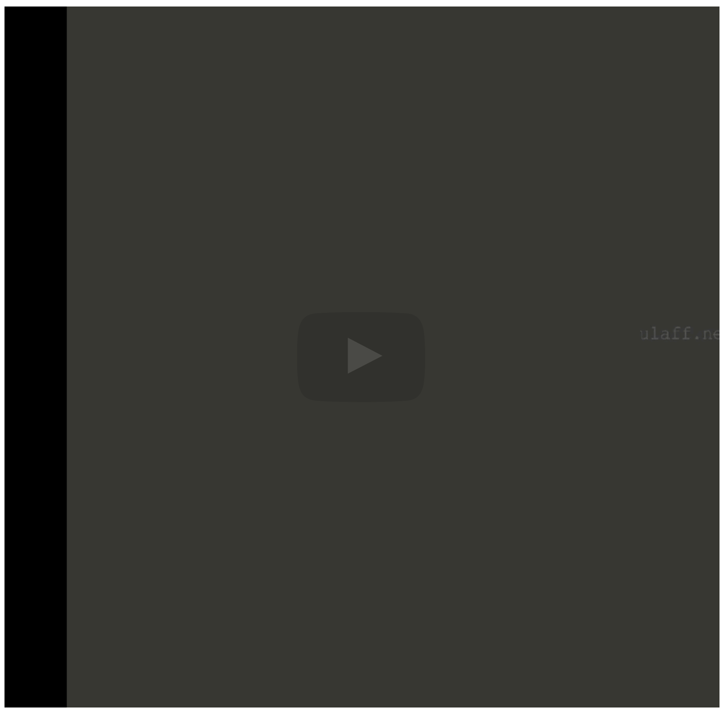

4:11 / 4:11 2.0x

that we

can expand our argument to yet another element?" And then an inductive argument

would allow us to show it for the entire linear system, for any matrix L that's

n by n. Alright, that's the approach. Now what goes into computing chi_1? Well,

in exact arithmetic chi_1 would be computed as psi_1 minus l_10 transpose

times x_0. And then the result of that would be divided by lambda_11.

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Reading assignment

1.0/1.0 point (graded)

Read Unit 6.4.1 of the notes. [LINK]

- [x] done

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Homework 6.4.1.1

1/1 point (graded)

1/1 point (graded)
Prove Lemma 6.4.1.2.

Hint: Reason the cases where $k = 0$ and $k = 1$ separately from the case where $k > 1$.

- [x] done

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Correct (1/1 point)

Homework 6.4.1.2

1/1 point (graded)
Prove Theorem 6.4.1.3 for the case where $n = 1$.

- [x] done

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Homework 6.4.1.3

1/1 point (graded)
Prove Theorem 6.4.1.3 for the case where $n = 2$.

- [x] done

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Homework 6.4.1.4

1/1 point (graded)
Prove Theorem 6.4.1.3 for the case where $n \geq 1$.

- [x] done

Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

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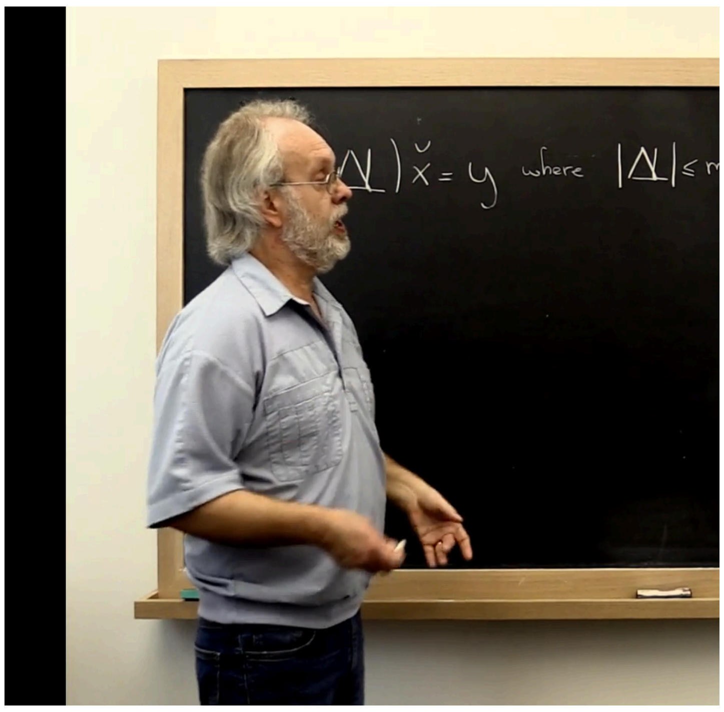

error result. Okay the computed solution to the lower

triangular system is the exact solution to a slightly changed matrix, a slightly

changed system where matrix L has been changed slightly. And we could put it

bound on that element-wise, because element-wise is less than or equal to

the max of gamma_2, gamma_n-1 times the absolute value of L. Okay? Now, since we

kind of relaxed bounds all along in order to come up

0:46 / 2:22

2.0x

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